

National University of Computer and Emerging Sciences, Lahore Campus

	Course Name:	Applied Calculus	Course Code:	MT-1001
	Program:	BS EE/Robotics	Semester:	Fall 2022
	Duration:	180 Minutes	Total Marks:	100
	Exam Date:	15-12-2022	Weight:	50
	Section:	ALL	Page(s):	2
	Exam Type:	Final		

Student Name: _____ Roll No. _____ Section: _____

- Instructions:
1. Use permanent ink pencil only. Any part done using soft pencil except sketching graphs will not be marked and cannot be claimed for rechecking.
 2. If need arises, make valid assumptions and clearly mention it with your answer.

Question 1.

[CLO 02, 20 Marks]

- What is the domain and range of $y = \frac{x}{|x|}$. Is the function even or odd? 
- Sketch the graph of $y = -\frac{1}{2}|x - 2| + 7$ by applying appropriate transformations.
- Using the concept of limits, find the vertical and/or horizontal asymptotes of $f(x) = \frac{8}{x^2 - 4}$, if any. $(x = \pm 2)$ (∞)
- At what points does the function $f(x) = \frac{x+1}{x^2 - 4x + 3}$ fail to be continuous? At which points, if any, are the discontinuities removable? $x = 3, x = 1$

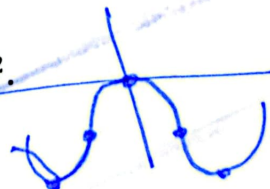
Question 2.

[CLO 03, 20 Marks]

(a) The voltage V (volts), current I (amperes), and resistance R (ohms) of an electric circuit are related by the equation $V = IR$. Suppose that V is increasing at the rate of 1 volt/sec while I is decreasing at the rate of $\frac{1}{3}$ amp/sec. Let t denote time in seconds.

- What is the value of $\frac{dV}{dt}$? 1 V/s
- What is the value of $\frac{dI}{dt}$? $-\frac{1}{3} \text{ A/s}$
- What equation relates $\frac{dR}{dt}$ to $\frac{dV}{dt}$ and $\frac{dI}{dt}$? $\frac{dR}{dt} = \frac{I \cdot \frac{dV}{dt} - V \cdot \frac{dI}{dt}}{I^2}$
- Find the rate at which R is changing when $V = 12$ volts and $I = 2$ amp. Is R increasing, or decreasing? $R = \frac{1}{2} \Omega/s$

(b) Using differentiation, sketch the graph of $y = x^4 - 2x^2$.



Question 3.

[CLO 04, 10 Marks]

Sketch the region enclosed by the given curves. Decide whether to integrate with respect to x or y . Then find the area of the region.

$$\frac{2(12-4x)^{3/2}}{3} - \frac{x^2}{2}$$

$$4x + y^2 = 12, y = x$$

Question 4.

(a) Evaluate the integral

$$\int_0^1 \frac{1}{\sqrt{1-x^2}} dx$$

$$\frac{\pi}{2}$$

(b) Show that whether the given integral is Divergent or Convergent.

$$\int_0^{\infty} \pi^{-y^2} dy$$

$$\infty$$

Question 5.

a) Find the parametric equation for the line through $(0, -7, 0)$ perpendicular to the plane $x + 2y + 2z = 13$.

b) Determine whether the lines

$$L_1: x = 5 - 12t, y = 3 + 9t, z = 1 - 3t \text{ and } L_2: x = 3 + 8s, y = -6s, z = 7 + 2s$$

are parallel, skew, or intersecting. If they intersect, find the point of intersection.

c) Find the distance between the planes $2x + 5y - z = 8$ and $2x + 5y - z = 5$.

d) Find the angle between the planes $x + y = 1$ and $2x + y - 2z = 2$.

e) Find equation for the plane through $P(1, -1, 2)$, $Q(2, 1, 3)$ and $R(-1, 2, -1)$.

f) Find an equation of the plane that contains the line $x = 1 + t, y = 2 - t, z = 4 - 3t$ and is parallel to the plane $5x + 2y + z = 1$.